Experiment Number:01

Experiment Name: Introduction to Logisim evaluation and implementation of basic logic gates.

Introduction to Logisim Evaluation :

Description:

Logisim is a digital circuit simulation software used to design and test logic circuits. In this experiment, basic logic gates such as AND, OR, NOT, NAND, NOR, XOR, and XNOR are implemented by placing the required gate components from the library. Input pins are used to provide binary values (0 or 1), and output pins display the resulting output according to each gate's truth table. By changing the input values, the behavior of each logic gate can be observed and verified. This experiment helps in understanding the working principles of digital logic circuits and provides practical experience in circuit design and simulation.

Theory:

Logic gates are the fundamental building blocks of digital electronics. They perform logical operations on one or more binary inputs (0 and 1) to produce a single binary output. Each logic gate follows a specific Boolean expression and truth table. The most common logic gates are AND, OR, NOT, NAND, NOR, XOR, and XNOR. These gates are used in digital systems such as computers, calculators, communication devices, and control circuits. Logisim provides a virtual platform to design, simulate, and verify the operation of these logic gates without using physical hardware.

By using this website:

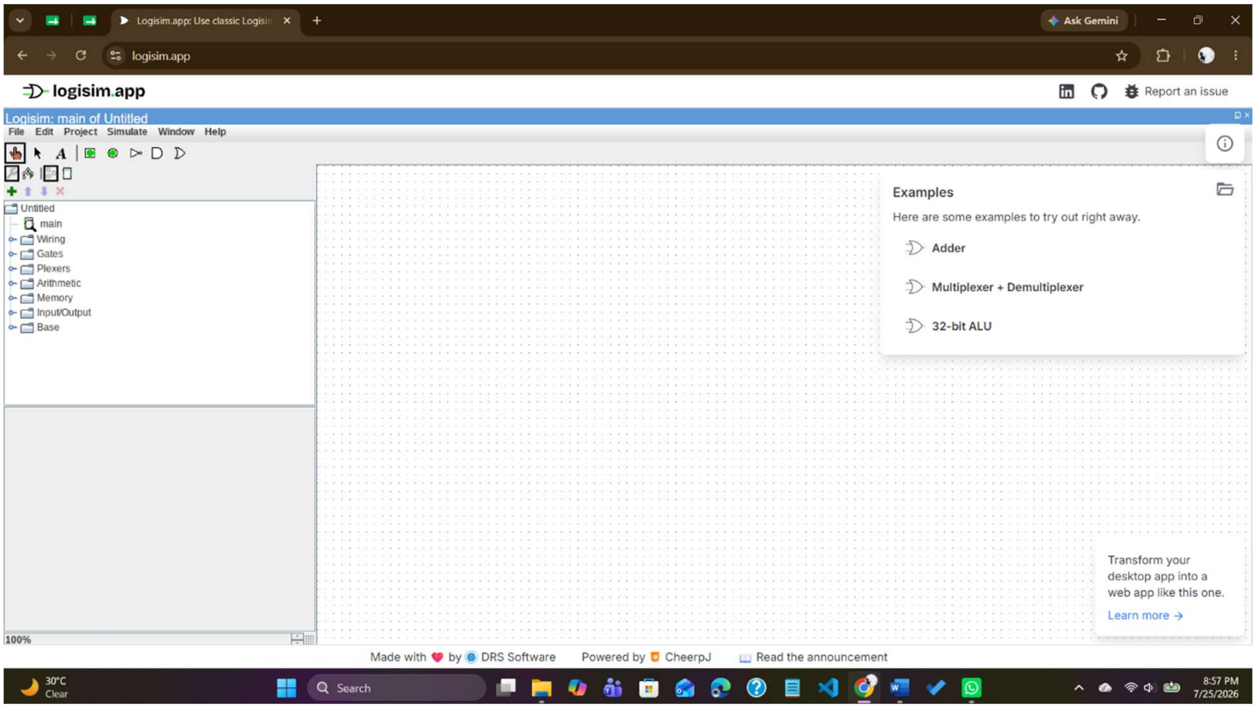

Fig 01:Using Logisim website

Introduction to Basic Logic Gates:

AND with NAND:

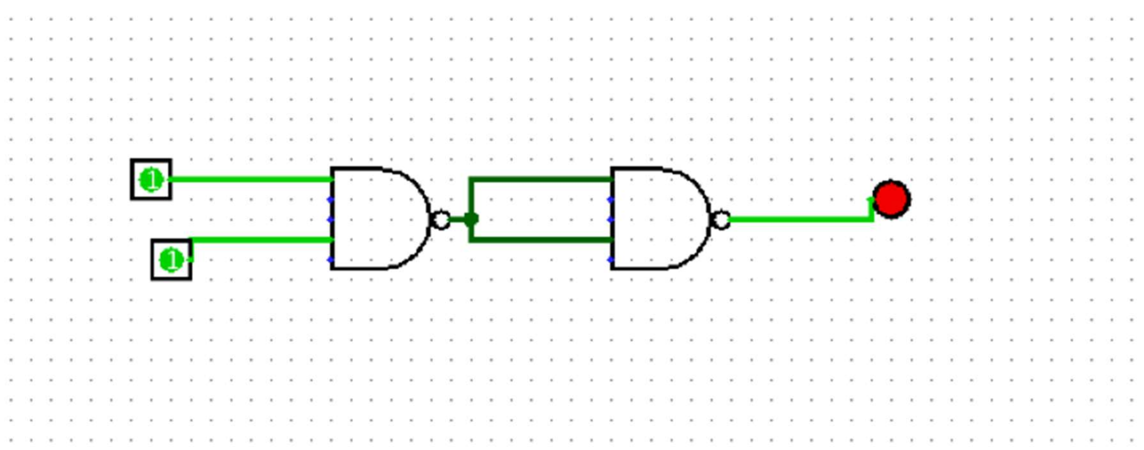

Fig 02:AND gate implementation with NAND

AND with NOR:

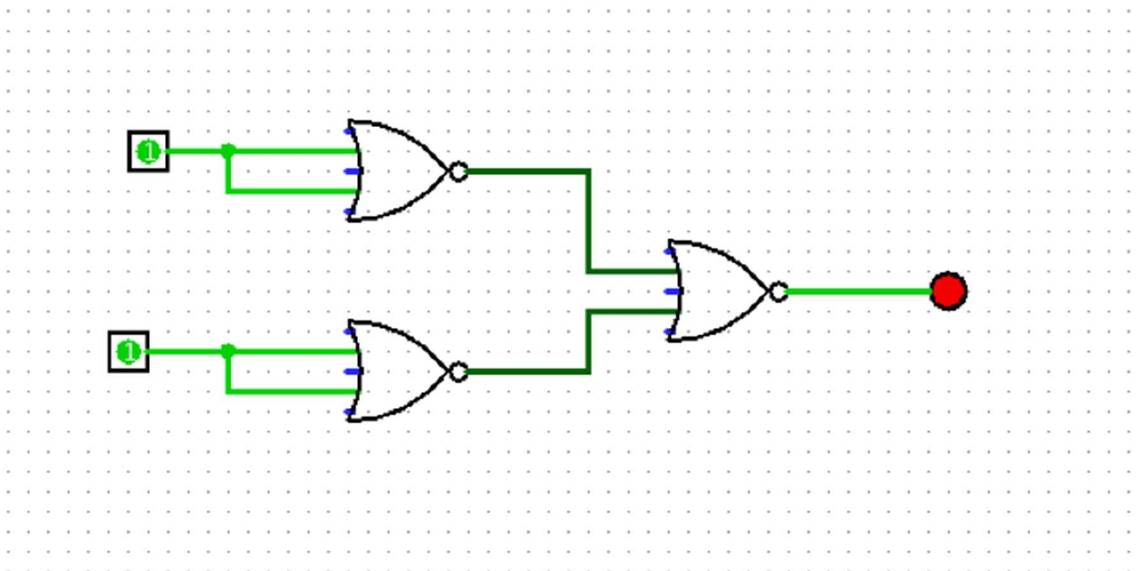

Fig 03:AND gate implementation with NOR

OR with NAND:

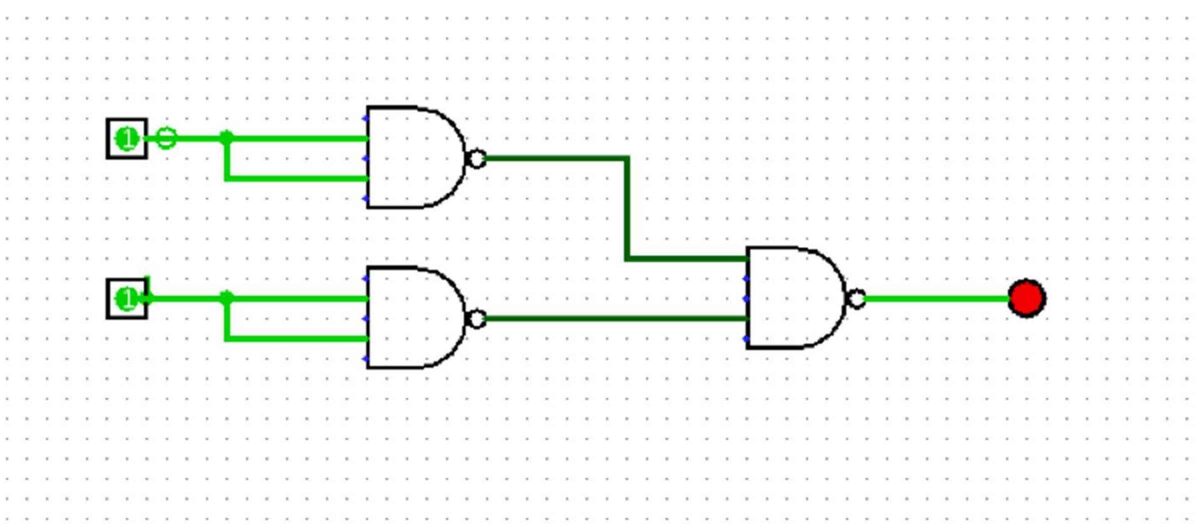

Fig 04:OR gate implementation with NAND

OR with NOR:

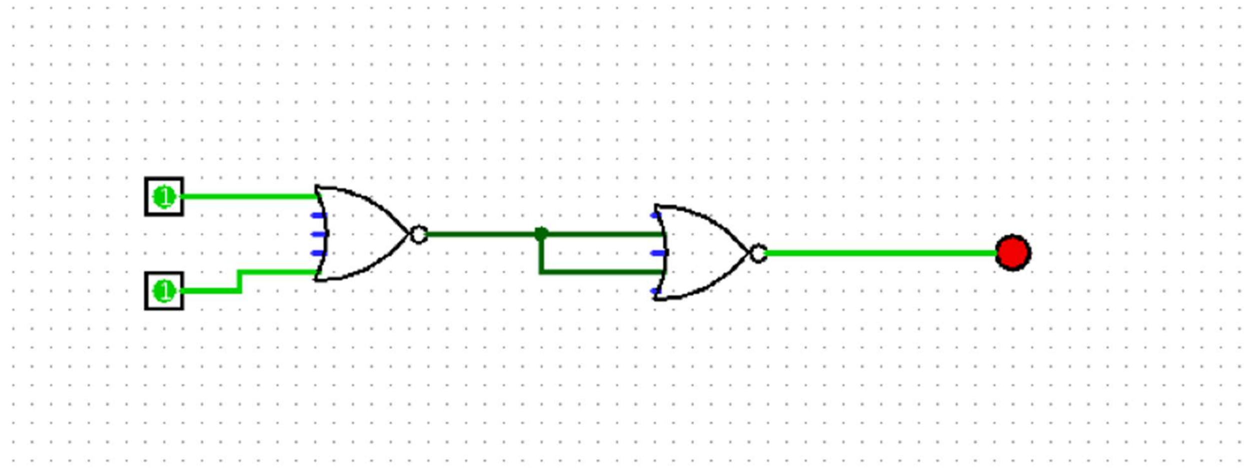

Fig 05:OD gate implementation with NOR

NOT with NAND:

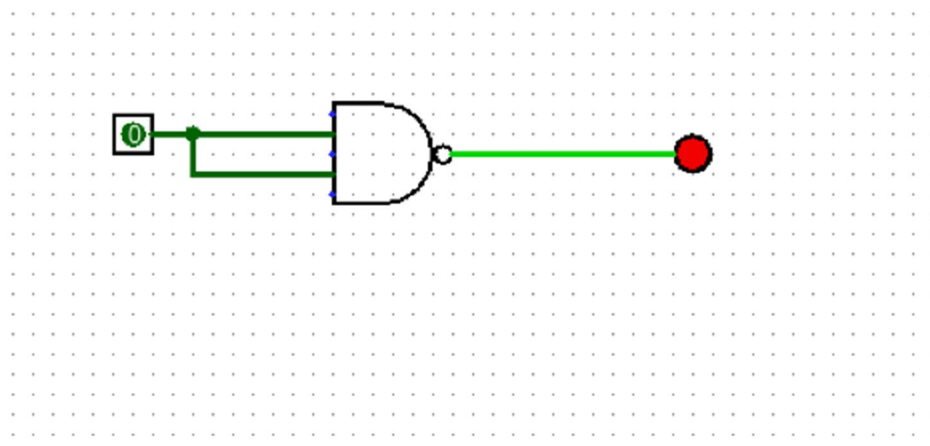

Fig 06:NOT gate implementation with NAND

NOT with NOR:

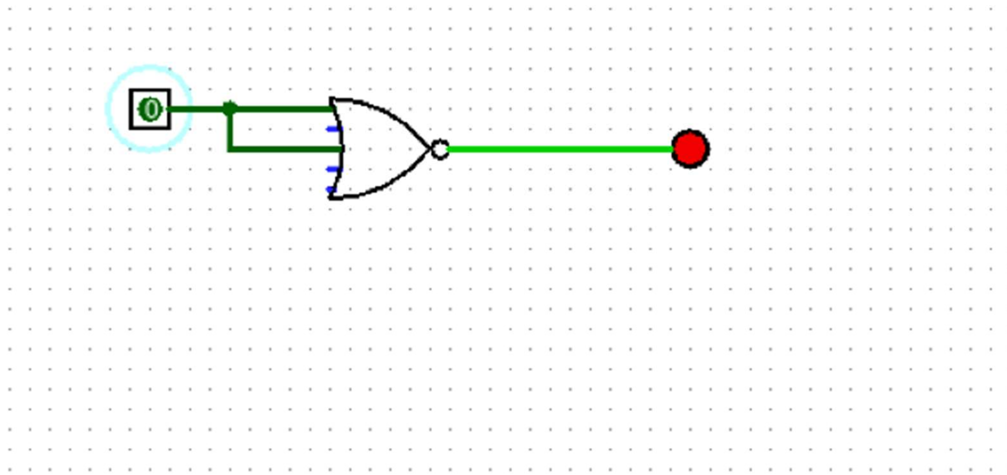

Fig 07:NOT gate implementation with NOR

Circuit of Full-Adder using Logic Gates:

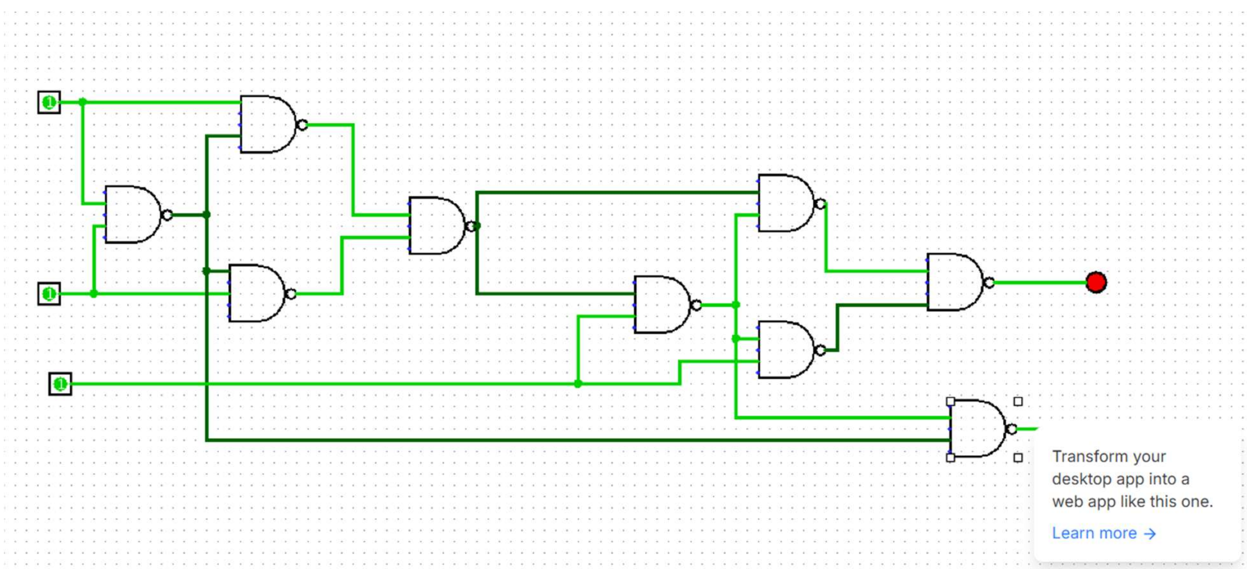

Fig 08:Full-Adder Circuit implementation with NAND